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# Computer Science & Information Technology

## Discrete Mathematics

DPP: 1

## Set Theory and Algebra

**Q1** Which of the following statements are true?

I. $\phi \in \phi$

II. $\phi \subset \phi$

III. $\phi \subseteq \phi$

IV. $\phi \in \{\phi\}$

V. $\phi \subset \{\phi\}$

Vi. $\phi \subseteq \{\phi\}$

**Q2** If a set A has 63 proper subsets, then what is the cardinality of A?

**Q3** If a set A has 64 subsets of odd cardinality, then what is |A|?

(A) 6 (B) 63

(C) 7 (D) 128

**Q4** How many subset of {1, 2, 3, ..., 11} contain at-least one even integer?

**Q5** Let A = {1, 2, 3, 4, 5, 7, 8, 10, 11, 14, 17, 18}
How many subsets of A contain six elements?

**Q6** Let A = {2, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 15}
How many six-elements subsets of A contain four even integers and two odd integers?

**Q7** Let A = {1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12}
How many subsets of A contain only odd integers?

**Q8** Suppose A, B, C and D are subsets of U (the universe) with A as a subset of B and C as subset of D
i.e A ⊆B and C ⊆ D, then consider the following statements

**I.** A ∩ C ⊆ B ∩ D

**II.** A ∪ C ⊆ B ∪ D

Which of the following is correct options?

(A) Only I is true

(B) Only II is true

(C) Neither I nor II is true

(D) Both I and II are true

**Q9** Let A = {1, 2, 3, ... 15}. How many subsets of A contains all of the odd integers in A?

**Q10** Let $A, B \subseteq$, where $A = \{x \mid x^2 - 7x = -12\}$ and $B = \{x \mid x^2 - x = 6\}$. Determine $A \cup B$ and $A \cap B$.

(A) $A \cup B$ = {5} and $A \cap B$ = {−2, 3, 4}

(B) $A \cup B$= {3} and $A \cap B$ = {−2, 3, 4}

(C) $A \cup B$ = {−2, 3, 4} and $A \cap B$ = {3}

(D) $A \cup B$ = {2, 3, 4} and $A \cap B$= {5}

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# Answer Key

| | |
|---|---|
| Q1 | 4~4 |
| Q2 | 6~6 |
| Q3 | (C) |
| Q4 | 1984~1984 |
| Q5 | 924~924 |
| Q6 | 225~225 |
| Q7 | 63~63 |
| Q8 | (D) |
| Q9 | 128~128 |
| Q10 | (C) |

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# Hints & Solutions

**Q1 Text Solution:**

I. $\phi \in \phi$ is false. The empty set has no members.

II. $\phi \subset \phi$ is false. The empty set is not a proper subset of itself.

III. $\phi \subseteq \phi$ is true. The empty set is a subset of every set

$\therefore$ subset of itself

IV. $\phi \in \{\phi\}$is true. phi is a member here.

V. $\phi \subset \{\phi\}$ is true. The empty set is a proper subset of itself.

VI. $\phi \subseteq \{\phi\}$ is true. The empty set is a subset of every set.

**Q2 Text Solution:**

If a set has n elements then the number of subsets will be $2^n$ and the number of proper subsets will be $2^n - 1$.

A has 63 proper subsets, so $2^n - 1 = 63$

$2^n = 63 + 1$

$2^n = 64$

$2^n = 2^6$

$\therefore n = 6$

The cardinality of A is 6

**Q3 Text Solution:**

The number of subsets for {1, 2, 3, ... n} with odd cardinality is $2^{n-1}$.

Number of subsets with cardinality i = ${}^nC_i$

So, the number of subsets with odd cardinality

$\sum i$ = 1, 3 ... n – 1 ${}^nC_i = 2^{n-1}$.

Now, given

$2^{n-1} = 64$

$2^{n-1} = 2^6$

$n - 1 = 6$

[Bases are same, so equating power]

$n = 7$

**Q4 Text Solution:**

$2^{11}$ subset for {1, 2, 3, ....11}

$2^6$ subset for {1, 3, 5, 7, 9, 11} contains none of the even integers {2, 4, 6, 8, 10}.

Hence, there are $2^{11} - 2^6 = 1984$ subsets that contain at least one even integer.

**Q5 Text Solution:**

If we choose 6 elements from a set of 12 elements where order does not matter. Then we can do it in ${}^{12}C_6$ ways.

For example consider a set = {1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12} out of this set we have to choose a subset of 6 elements. This can be done in following ways:

1, 2, 3, 4, 5, 6}, {2, 3, 4, 5, 6, 7},{3, 4, 5, 6, 7, 8}, {4, 7, 8, 3, 11, 12}, {5, 7, 9, 10, 11, 12}..... and so on.

That means the arrangement or order of elements does not matter, therefore we can do it using combinations.

$${}^{12}C_6 = \frac{12 \times 11 \times 10^{2} \times 9^{3} \times 8^{2} \times 7 \times 6!}{6! \times 6 \times 5 \times 4 \times 3 \times 2 \times 1}$$

$= 11 \times 12 \times 7$

$= 77 \times 12$

$= 924$

**Q6 Text Solution:**

Out of 6 element subsets, we can choose 4 even integers in ${}^6C_4$ ways.

Similarly to find 2 odd integers out of 6 element subset, can be done in ${}^6C_2$ ways.

Therefore

$${}^6C_4 * {}^6C_2 = \frac{6^{3} \times 5 \times 4!}{2 \times 4!} \times \frac{6^{3} \times 5 \times 4!}{4! \times 2}$$

$= 15 * 15 = 225$

**Q7 Text Solution:**

In the given set, there are 6 odd integers, we have two choices for each odd integer to be included or not included, therefore total possibilities = $2^6 - 1 = 63$.

**Q8 Text Solution:**

**I.** $A \cap C \subseteq B \cap D$, is True.

Let a be an arbitrary element of $A \cap C$, so $a \in A \cap C$ then $a \in A \subseteq B$, so $a \in B$ and $a \in C \subseteq D$, so $a \in D$. That concludes that $a \in B$ and $a \in D$, therefore by definition $a \in B \cap D$. If follows that

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every element of A ∩ C belongs to B ∩ D, which by definition means A ∩ C ⊆ B ∩ D.

**II.** A ∪ C ⊆ B ∪ D, is True.

If a is an arbitrary element that belongs to A ∪ C then it definitely belongs to B ∪ D as A ⊆ B and C ⊆D.

**Q9 Text Solution:**

In the given set A = {1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15}

There are 8 odd integers. For all odd integer we have choices whether to include it or not with the 7 even integers in the set.

Therefore possiblities = $2^7$ = 128.

**Q10 Text Solution:**

$x^2-7x=-12 \Rightarrow x^2-7x+12=0 \Rightarrow (x-4)(x-3)=0 \Rightarrow x=4, x=3.$

$x^2-x=6 \Rightarrow x^2-x-6=0 \Rightarrow (x-3)(x+2)=0 \Rightarrow x=3, x=-2.$

Consequently, $A \cap B = \{3\}$ and $A \cup B = \{-2, 3, 4\}$.

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